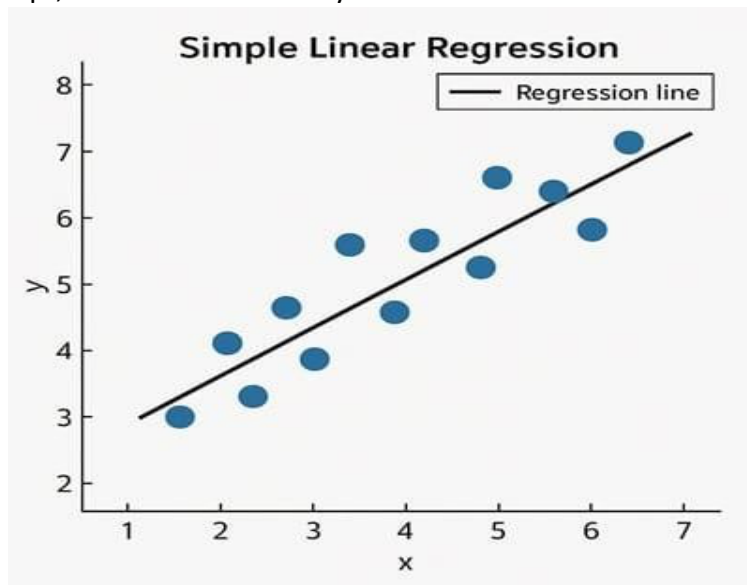


Algorithms

1. Linear Regression

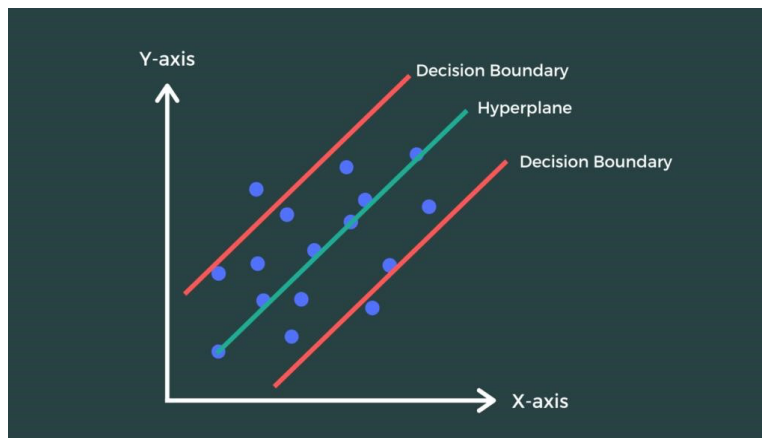
Description:

Linear Regression is a simple supervised learning algorithm used for predicting continuous values. It finds a straight-line relationship between input features and output. In house price prediction, it estimates price based on features like bedrooms, sqft, etc. It is fast and easy but works best when data has a linear pattern.



2. Support Vector Regression (SVR)

Description:

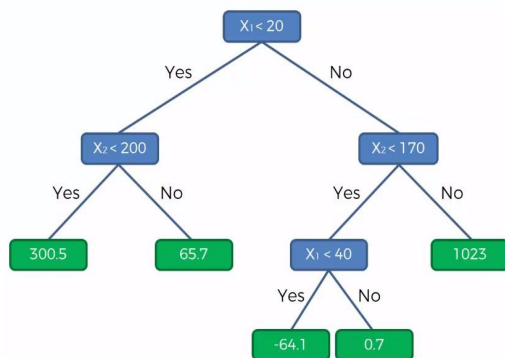


SVR is the regression version of Support Vector Machine. It tries to fit the best line within a margin (called epsilon tube) while minimizing error. It is powerful for complex data but sensitive to scaling. In this project, it helps model non-linear relationships between house features and price.

Decision Tree Regressor

Description:

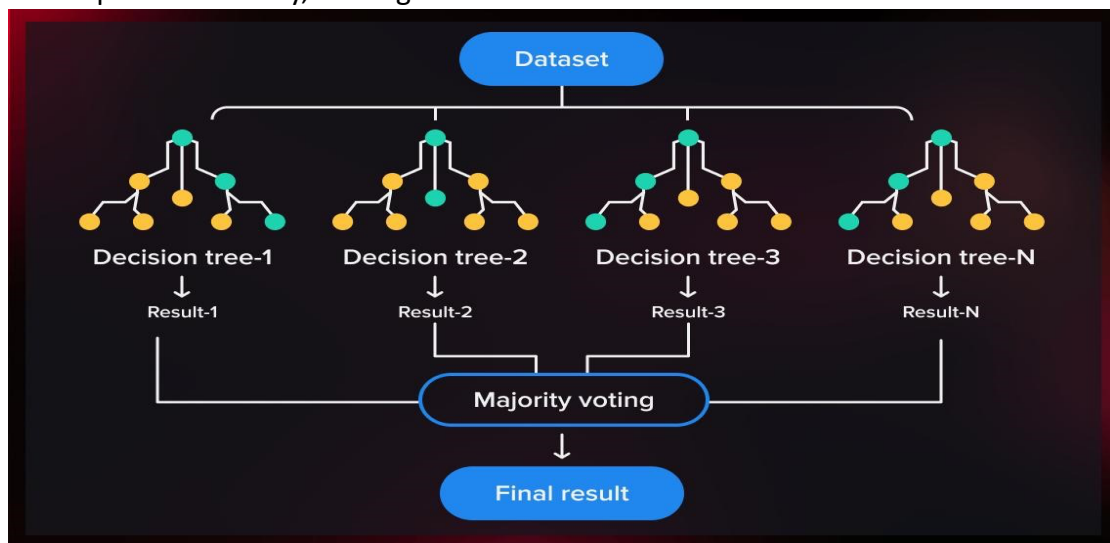
A Decision Tree Regressor splits data into branches based on feature values. Each split reduces error and leads to a final prediction at the leaf node. It is easy to understand but may overfit if the tree becomes too deep.



Random Forest Regressor

Description:

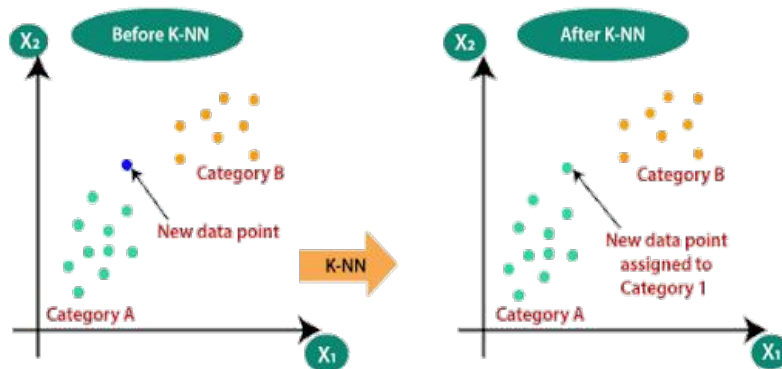
Random Forest is an ensemble method that combines multiple decision trees. Each tree gives a prediction, and the final output is the average of all. It reduces overfitting and improves accuracy, making it one of the best models for this dataset.



K-Nearest Neighbors (KNN) Regressor

Description:

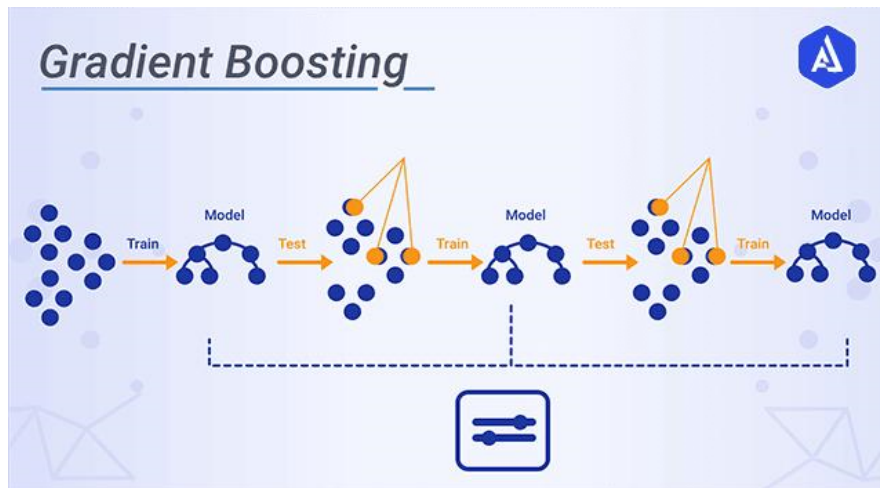
KNN predicts house price based on the average of nearest similar houses. It uses distance (like Euclidean distance) to find similar data points. It is simple but slower for large datasets.



3. Gradient Boosting Regressor

Description:

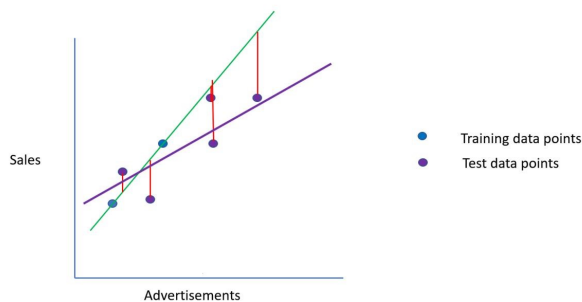
Gradient Boosting builds models sequentially, where each new model corrects errors of the previous one. It gives very high accuracy and is widely used in real-world applications. It performs very well in house price prediction.



4. Ridge Regression

Description:

Ridge Regression is an improved version of Linear Regression. It adds a penalty (regularization) to reduce overfitting. It works well when features are correlated and improves model stability.



Final Conclusion:

This project was done to predict house prices using different machine learning algorithms and house details like bedrooms, bathrooms, and square footage. Several models were tested to see which one gives more accurate predictions. Simple models like Linear Regression work well for basic predictions, while advanced models like Random Forest and Gradient Boosting give better results by learning patterns from the data more effectively. These advanced models also help reduce prediction errors. Overall, the project shows that ensemble models are more reliable for predicting house prices accurately.

What I Learned:

- I learned how machine learning can help in predicting house prices using house details.
- I understood how different regression models work and how their results can change.
- I learned why preparing the data properly is important before training a model.
- I understood how to check which model performs better using accuracy measures.
- I learned that models like Random Forest and Gradient Boosting usually give better predictions.
- Overall, this project gave me practical experience in building and testing machine learning models.